

1. Random Processes

1.1 Definition and Classification of Processes

In the study of deterministic signals, we often encounter four types or classes of signals:

- (1) *Continuous time and continuous amplitude signals* are a function of a continuous independent variable, time. The range of the amplitude of the function is also continuous.
- (2) *Continuous time and discrete amplitude signals* are a function of a continuous independent variable, time—but the amplitude is discrete.
- (3) *Discrete time and continuous amplitude signals* are functions of a quantized or discrete independent time variable, while the range of amplitudes is continuous.
- (4) *Discrete time and discrete amplitude signals* are functions where both the independent time variable and the amplitude are discrete.

1.2 Random Process

A random process (or stochastic process) is a collection of random variables indexed by time:

$$X(t), \quad t \in T \tag{1}$$

For each fixed t , $X(t)$ is a random variable.

1.3 Cumulative Distribution Function (CDF)

The CDF of a random process at time t is defined as:

$$F_X(x, t) = P(X(t) \leq x) \tag{2}$$

Properties:

- $0 \leq F_X(x, t) \leq 1$

- $F_X(x, t)$ is non-decreasing in x
- $\lim_{x \rightarrow \infty} F_X(x, t) = 1$

Since a continuous random variable will typically have a zero probability of taking on a specific value, we avoid talking about such probabilities. Instead, events of the form $\{X \leq x\}$ can be considered.

1.3.1 Definition

The CDF of a random variable, X , is

$$F_X(x) = \Pr(X \leq x). \quad (3)$$

From this definition, several properties of the CDF can be inferred. First, since the CDF is a probability, it must take on values between 0 and 1. Since random variables are real valued, it is easy to conclude that $F_X(-\infty) = 0$ and $F_X(\infty) = 1$. That is, a real number cannot be less than $-\infty$ and must be less than ∞ . Next, if we consider two fixed values, x_1 and x_2 , such that $x_1 < x_2$, then the event $\{X \leq x_1\}$ is a subset of $\{X \leq x_2\}$. Hence, $F_X(x_1) \leq F_X(x_2)$. This implies that the CDF is a monotonic nondecreasing function. Also, we can break the event $\{X \leq x_2\}$ into the union of two mutually exclusive events, $\{X \leq x_1\} \cup \{x_1 < X \leq x_2\}$. Hence,

$$F_X(x_2) = F_X(x_1) + \Pr(x_1 < X \leq x_2), \quad (4)$$

or equivalently,

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1). \quad (5)$$

Thus, the CDF can also be used to measure the probability that a random variable takes on a value in a certain interval. These properties of CDFs are summarized as follows:

1. $F_X(-\infty) = 0$, $F_X(\infty) = 1$,
2. $0 \leq F_X(x) \leq 1$,
3. For $x_1 < x_2$, $F_X(x_1) \leq F_X(x_2)$,
4. For $x_1 < x_2$, $\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$.

1.4 Probability Density Function (PDF)

For a continuous random process, the PDF is defined as:

$$f_X(x, t) = \frac{\partial}{\partial x} F_X(x, t) \quad (6)$$

Probability over an interval:

$$P(a \leq X(t) \leq b) = \int_a^b f_X(x, t) dx \quad (7)$$

Properties:

- $f_X(x, t) \geq 0$
- $\int_{-\infty}^{\infty} f_X(x, t) dx = 1$

While the CDF introduced in the last section represents a mathematical tool to statistically describe a random variable, it is often quite cumbersome to work with CDFs. For example, we will see later in this chapter that the most important and commonly used random variable, the Gaussian random variable, has a CDF which cannot be expressed in closed form. Furthermore, it can often be difficult to infer various properties of a random variable from its CDF. To help circumvent these problems, an alternative and often more convenient description known as the probability density function is often used.

1.4.1 Definition

The PDF of the random variable X evaluated at the point x is

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}. \quad (8)$$

As the name implies, the PDF is the probability that the random variable X lies in an infinitesimal interval about the point $X = x$, normalized by the length of the interval.

Note that the probability of a random variable falling in an interval can be written in terms of its CDF. For continuous random variables,

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x), \quad (9)$$

so that

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon} = \frac{dF_X(x)}{dx}. \quad (10)$$

Hence, it is seen that the PDF of a random variable is the derivative of its CDF.

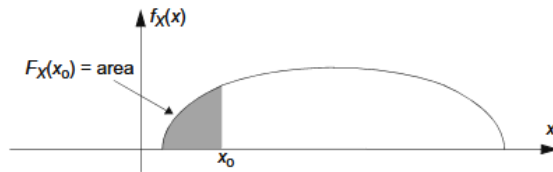


Figure 1: Relationship between the PDF and CDF of a random variable

Conversely, the CDF of a random variable can be expressed as the integral of its PDF.

From the definition of the PDF, it is apparent that the PDF is a nonnegative function although it is not restricted to be less than unity as with the CDF. From the properties of the CDFs, we can also infer several important properties of PDFs. Some properties of PDFs are:

1. $f_X(x) \geq 0$,
2. $f_X(x) = \frac{dF_X(x)}{dx}$,
3. $F_X(x) = \int_{-\infty}^x f_X(y) dy$,
4. $\int_{-\infty}^{\infty} f_X(x) dx = 1$,
5. $\int_a^b f_X(x) dx = \Pr(a < X \leq b)$.

1.5 Probability Mass Function (PMF)

For a discrete random process, the PMF is defined as:

$$p_X(x, t) = P(X(t) = x) \quad (11)$$

Properties:

- $0 \leq p_X(x, t) \leq 1$
- $\sum_x p_X(x, t) = 1$

1.6 Relation between CDF, PDF, and PMF

1.6.1 Relation with PDF (Continuous Case)

For a continuous random variable, the PDF is related to the CDF as:

$$f_X(x) = \frac{d}{dx} F_X(x) \quad (12)$$

Conversely, the CDF can be obtained from the PDF:

$$F_X(x) = \int_{-\infty}^x f_X(u) du \quad (13)$$

Probability over an interval:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx \quad (14)$$

1.6.2 Relation with PMF (Discrete Case)

For a discrete random variable, the PMF is:

$$p_X(x) = P(X = x) \quad (15)$$

The CDF is obtained by summation:

$$F_X(x) = \sum_{u \leq x} p_X(u) \quad (16)$$

Summary:

- CDF is the most general representation.
- PDF is the derivative of CDF (continuous case).
- PMF is the discrete probability representation.
- CDF is obtained by integration (continuous) or summation (discrete).

1.7 The Gaussian Random Variable

The Gaussian random variable is the most commonly used and of great importance. Many physical phenomena can be modeled as Gaussian random variables, including the thermal noise encountered in electronic circuits.

The PDF of a Gaussian random variable is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right) \quad (17)$$

where m is the mean, and σ^2 is the variance.

The special case when $m = 0$ and $\sigma = 1$ is called a **standard normal random variable**.

The CDF of a Gaussian random variable is

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy \quad (18)$$

1.8 Operations on a Single Random Variable

1.8.1 Expected Value of a Random Variable

The expected value of a random variable X which has a PDF $f_X(x)$ is

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx. \quad (19)$$

$$\mu_X = E[X] \quad (20)$$

The expected value is also called the average, mean, expectation, or first moment.

For discrete random variables,

$$E[X] = \sum_i x_i P_X(x_i). \quad (21)$$

Given a random variable X with PDF $f_X(x)$, the expected value of a function $g(X)$ is given by

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx. \quad (22)$$

For discrete random variables,

$$E[g(X)] = \sum_k g(x_k) P_X(x_k). \quad (23)$$

For any constants a and b ,

$$E[aX + b] = aE[X] + b. \quad (24)$$

$$E\left[\sum_{k=1}^N g_k(X)\right] = \sum_{k=1}^N E[g_k(X)]. \quad (25)$$

Mean of sum of functions = Sum of their individual means.

Moments

The n th moment of a random variable X is defined as

$$E[X^n] = \int x^n f_X(x) dx. \quad (26)$$

For discrete random variables,

$$E[X^n] = \sum_k (x_k)^n P_X(x_k). \quad (27)$$

- **Mean (First moment):**

$$\mu_X = \bar{X} = E[X] \quad (28)$$

- **Variance (Second central moment):**

$$\sigma_X^2 = E[(X - \mu_X)^2] \quad (29)$$

- **Skewness (Third central moment):**

$$c_s = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^3\right] \quad (30)$$

- **Kurtosis (Fourth central moment):**

$$c_k = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^4\right] \quad (31)$$

Pairs of Random Variables

Joint Probability Density Function

The joint PDF of random variables X and Y is defined as

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y). \quad (32)$$

The joint CDF is given by

$$F_{X,Y}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u, v) du dv. \quad (33)$$

Properties

1.

$$f_{X,Y}(x, y) \geq 0, P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = \int_{y_1}^{y_2} \int_{x_1}^{x_2} f_{X,Y}(x, y) dx dy \quad (34)$$

2.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1 \quad (35)$$

3.

$$F_{XY}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u, v) du dv \quad (36)$$

4.

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) \quad (37)$$

5. Marginal PDFs:

$$f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{x=-\infty}^{\infty} f_{X,Y}(x, y) dx \quad (38)$$

1.8.2 Conditional Probability:

The conditional PDF of X given $Y = y$ is defined as

$$P_{X|Y}(x|y) = \frac{P_{X,Y}(x, y)}{P_Y(y)} \quad (39)$$

The correlation between two random variables X and Y is defined as

$$R_{X,Y} = E[XY] = \int \int xy f_{X,Y}(x, y) dx dy. \quad (40)$$

Two random variables which have a correlation of zero are said to be **orthogonal**.

The covariance between two random variables X and Y is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]. \quad (41)$$

Equivalently,

$$\text{Cov}(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{X,Y}(x, y) dx dy. \quad (42)$$

If two random variables have a covariance of zero, then they are said to be **uncorrelated**.

$$\text{Cov}(X, Y) = R_{XY} - \mu_X \mu_Y \quad (43)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) \quad (44)$$

The correlation coefficient between two random variables X and Y is defined as

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}. \quad (45)$$

$$= \frac{\sigma_{XY}}{\rho_X, \rho_Y} \quad (46)$$

$$-1 \leq \rho_{XY} \leq 1 \quad (47)$$

If $\rho_{XY} = 0$, the random variables are **uncorrelated**.

$$\rho_{XY} = \pm 1 \quad (48)$$

implies that X and Y are **completely correlated**.

- Independence implies uncorrelatedness.
- Uncorrelatedness does not imply independence.

Independent Random Variables

Two random variables (or events) A and B are said to be statistically independent if

$$P(A, B) = P(A) P(B). \quad (49)$$

Equivalently,

$$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y). \quad (50)$$

Hence, the joint CDF factors as

$$F_{X,Y}(x, y) = F_X(x) F_Y(y). \quad (51)$$

Two random variables are statistically independent if their joint CDF factors into a product of the marginal CDFs.

For independent random variables, the joint PDF satisfies

$$f_{X,Y}(x, y) = f_X(x) f_Y(y). \quad (52)$$

Consequently, the conditional PDFs reduce to the marginals:

$$f_{X|Y}(x|y) = f_X(x), \quad f_{Y|X}(y|x) = f_Y(y). \quad (53)$$

1.9 Independent and Identically Distributed (IID) Random Variables

In many applications, we are able to observe an experiment repeatedly. Each new observation can occur with an independent realization of whatever random phenomena control the experiment. This type of situation gives rise to independent and identically distributed random variables.

A sequence of random variables $X_1, X_2, \dots, X_n$ is said to be IID if

- **Identically distributed:**

$$F_{X_i}(x) = F_X(x), \quad \forall i = 1, 2, \dots, n \quad (54)$$

- **Independent:**

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n F_{X_i}(x_i) \quad (55)$$

For continuous random variables, the CDFs can be replaced with PDFs:

$$P_{X_i}(x) = P_X(x), \quad \forall i = 1, 2, \dots, n \quad (56)$$

and

$$P_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P_{X_i}(x_i). \quad (57)$$

1.10 Mathematical Tools for Studying Random Processes

As with random variables, we can mathematically describe a random process in terms of a cumulative distribution function, a probability density function, or a probability mass function. In fact, given a random process, $X(t)$, which is sampled at some specified point in time, $t = t_k$, the result is a random variable, $X_k = X(t_k)$. This random variable can then be described in terms of its PDF, $f_X(x_k; t_k)$. Note that an additional time variable has been added to the PDF. This is necessary due to the fact that the PDF of the sample of the random process may depend on when the process is sampled. If desired, the CDF or PMF can be used rather than the PDF to describe the sample of the random process.

The PDF (or CDF or PMF) of a sample of a random process taken at an arbitrary point in time goes a long way toward describing the random process, but it is not a complete description. To see this, consider two samples, $X_1 = X(t_1)$ and $X_2 = X(t_2)$, taken at two arbitrary points in time. The PDF, $f_X(x; t)$, describes both X_1 and X_2 , but

it does not describe the relationship between X_1 and X_2 . For some random processes, it might be reasonable to expect that X_1 and X_2 would be highly correlated if t_1 is near t_2 , while X_1 and X_2 might be virtually uncorrelated if t_1 and t_2 are far apart. To characterize relationships of this sort, a joint PDF of the two samples would be needed. Hence, it would be necessary to construct a joint PDF of the form

$$f_{X_1, X_2}(x_1, x_2; t_1, t_2). \quad (58)$$

This is referred to as a *second order PDF* of the random process $X(t)$.

Continuing with this reasoning, in order to completely describe the random process, it is necessary to specify an n th order PDF for an arbitrary n . That is, suppose random process is sampled at time instants $t_1, t_2, \dots, t_n$, producing the random variables $X_1 = X(t_1), X_2 = X(t_2), \dots, X_n = X(t_n)$. The joint PDF of the n samples,

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n), \quad (59)$$

for an arbitrary n and arbitrary sampling times will give a complete description of the random process. In order to make this notation more compact, the vectors

$$\mathbf{X} = (X_1, X_2, \dots, X_n)^T, \quad \mathbf{x} = (x_1, x_2, \dots, x_n)^T, \quad \mathbf{t} = (t_1, t_2, \dots, t_n)^T \quad (60)$$

are introduced and the n th order joint PDF is written as $f_{\mathbf{X}}(\mathbf{x}; \mathbf{t})$.

Unfortunately, for many realistic random processes, the prospects of writing down an n th order PDF is rather daunting. One notable exception is the Gaussian random process. However, for most other cases, specifying a joint PDF of n samples may be exceedingly difficult, and hence it is necessary to resort to a simpler but less complete description of the random process. The simplest is the mean function of the process.

1.11 Mean Function

The *mean function* of a random process is simply the expected value of the process. For continuous time processes this is written as

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x f_X(x; t) dx, \quad (61)$$

while for discrete time processes, the following notation is used:

$$\mu_X[n] = E[X[n]] = \int_{-\infty}^{\infty} x f_X(x; n) dx. \quad (62)$$

In general, the mean of a random process may change with time, but in many cases, this function is constant. Also, it is noted that only the first order PDF of the process is

needed to compute the mean function.

1.12 Autocorrelation Function

The *autocorrelation function*, $R_{XX}(t_1, t_2)$, of a continuous time random process, $X(t)$, is defined as the expected value of the product $X(t_1)X(t_2)$:

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X_1, X_2}(x_1, x_2; t_1, t_2) dx_1 dx_2. \quad (63)$$

For discrete time processes, the autocorrelation function is

$$R_{XX}[n_1, n_2] = E[X[n_1]X[n_2]] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X_1, X_2}(x_1, x_2; n_1, n_2) dx_1 dx_2. \quad (64)$$

Naturally, the autocorrelation function describes the relationship (correlation) between two samples of a random process. This correlation will depend on when the samples are taken; thus, the autocorrelation function is, in general, a function of two time variables. Quite often we are interested in how the correlation between two samples depends on how far apart the samples are spaced. To explicitly draw out this relationship, define a time difference variable, $\tau = t_2 - t_1$, and the autocorrelation function can then be expressed as

$$R_{XX}(t, t + \tau) = E[X(t)X(t + \tau)], \quad (65)$$

where we have replaced t_1 with t to simplify the notation even further.

1.13 The autocovariance function

The autocovariance function, $C_{XX}(t_1, t_2)$, of a continuous time random process, $X(t)$, is defined as the covariance of $X(t_1)$ and $X(t_2)$:

$$C_{XX}(t_1, t_2) = \text{Cov}(X(t_1), X(t_2)) = E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))]. \quad (66)$$

The definition is easily extended to discrete time random processes.

As with the covariance function for random variables, the autocovariance function can be written in terms of the autocorrelation function and the mean function:

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2). \quad (67)$$

Once the mean and autocorrelation functions of a random process have been computed, the autocovariance function is trivial to find.

The autocovariance function is helpful when studying random processes that can be represented as the sum of a deterministic signal, $s(t)$, plus a zero-mean noise process,

$N(t)$. If $X(t) = s(t) + N(t)$, then the autocorrelation function of $X(t)$ is

$$R_{XX}(t_1, t_2) = E[(s(t_1) + N(t_1))(s(t_2) + N(t_2))] = s(t_1)s(t_2) + R_{NN}(t_1, t_2), \quad (68)$$

using the fact that $\mu_N(t) = 0$. If the signal is strong compared to the noise, the deterministic part will dominate the autocorrelation function, and hence $R_{XX}(t_1, t_2)$ will not tell us much about the randomness in the process $X(t)$. On the other hand, the autocovariance function is

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - s(t_1)s(t_2) = R_{NN}(t_1, t_2) = C_{NN}(t_1, t_2). \quad (69)$$

Hence, the autocovariance function allows us to isolate the noise that is the source of randomness in the process.

1.14 Cross Correlation Function

For a pair of random processes $X(t)$ and $Y(t)$, the crosscorrelation function is defined as

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]. \quad (70)$$

Likewise, the cross-covariance function is

$$C_{XY}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(Y(t_2) - \mu_Y(t_2))]. \quad (71)$$

Stationary and Ergodic Random Processes

From the few simple examples given in the preceding section, we conclude that the mean function and the autocorrelation (or autocovariance) function can provide information about the temporal structure of a random process. We will delve into the properties of the autocorrelation function in more detail later in the chapter, but first the concepts of *stationarity* and *ergodicity* must be introduced.

1.14.1 Strict Sense Stationary Process

A continuous time random process $X(t)$ is *strict sense stationary* if the statistics of the process are invariant to a time shift. Specifically, for any time shift τ and any integer $n \geq 1$,

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; t_1 + \tau, t_2 + \tau, \dots, t_n + \tau) \quad (72)$$

In general, it is quite difficult to show that a random process is strict sense stationary since to do so, one needs to be able to express the general n th order PDF. On the other

hand, to show that a process is not strict sense stationary, one needs to show only that one PDF of any order is not invariant to a time shift. One example of a process that can be shown to be stationary in the strict sense is an IID process. That is, suppose $X(t)$ is a random process that has the property that $X(t)$ has an identical distribution for any t and that $X(t_1)$ and $X(t_2)$ are independent for any $t_1 \neq t_2$. In this case,

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = \prod_{i=1}^n f_{X_i}(x_i; t_i) \quad (73)$$

Since the n th order PDF is the product of first order PDFs and the first order PDF is invariant to a time shift, then the n th order PDF must be invariant to a time shift.

1.14.2 Wide Sense Stationary Process (WSS)

A random process is *wide sense stationary* (WSS) if the mean function and autocorrelation function are invariant to a time shift. In particular, this implies that

$$\mu_X(t) = \mu_X = \text{constant} \quad (74)$$

$$R_{XX}(t, t + \tau) = R_{XX}(\tau) \quad (75)$$

All strict sense stationary random processes are also WSS, provided that the mean and autocorrelation function exist. The converse is not true. A WSS process does not necessarily need to be stationary in the strict sense. We refer to a process that is not WSS as *nonstationary*.

Many of the processes we deal with are WSS and hence have a constant mean function and an autocorrelation function that depends only on a single time variable. Hence, in the remainder of the text, when a process is known to be WSS or if we are assuming it to be WSS, then we will represent its autocorrelation function by $R_{XX}(\tau)$. If a process is nonstationary or if we do not know if the process is WSS, then we will explicitly write the autocorrelation function as a function of two variables, $R_{XX}(t, t + \tau)$. For example, if we say that a process has a mean function of $\mu_X = 1$ and an autocorrelation function, $R_{XX}(\tau) = \exp(-|\tau|)$, then the reader can infer that the process is WSS, even if it is not explicitly stated.

1.14.3 Ergodicity

In order to calculate the mean or autocorrelation function of a random process, it is necessary to perform an ensemble average. In many cases, this may not be possible as we may not be able to observe all realizations (or a large number of realizations) of a random process. In fact, quite often we may be able to observe only a single realization. This would occur in situations where the conditions of an experiment cannot be duplicated

and hence the experiment is not repeatable. Is it possible to calculate the mean and/or autocorrelation function from a single realization of a random process? The answer is, sometimes, depending on the nature of the process.

To start with, consider the mean. Suppose a WSS random process $X(t)$ has a mean μ_X . We are able to observe one realization of the random process, $x(t)$, and wish to try to determine μ_X from this realization. One obvious approach would be to calculate the time average of the realization:

$$\langle x(t) \rangle = \lim_{t_0 \rightarrow \infty} \frac{1}{2t_0} \int_{-t_0}^{t_0} x(t) dt \quad (76)$$

However, it is not obvious if the time average of one realization is necessarily equal to the ensemble average. If the two averages are the same, then we say that the random process is *ergodic in the mean*.

One could take the same approach for the autocorrelation function. Given a single realization, $x(t)$, form the time-average autocorrelation function:

$$\mathfrak{R}_{xx}(\tau) = \langle x(t)x(t+\tau) \rangle = \lim_{t_0 \rightarrow \infty} \frac{1}{2t_0} \int_{-t_0}^{t_0} x(t)x(t+\tau) dt \quad (77)$$

If $\mathfrak{R}_{xx}(\tau) = R_{XX}(\tau)$ for any realization, $x(t)$, then the random process is said to be *ergodic in the autocorrelation*.

1.14.4 Ergodic Process Definition

A WSS random process is *ergodic* if ensemble averages involving the process can be calculated using time averages of any realization of the process. Two limited forms of ergodicity are

1. ergodic in the mean: $\langle x(t) \rangle = E[X(t)]$
2. ergodic in the autocorrelation: $\langle x(t)x(t+\tau) \rangle = E[X(t)X(t+\tau)]$

1.15 Properties of the Autocorrelation Function

Since the autocorrelation function, along with the mean, is considered to be a principal statistical descriptor of a WSS random process, we will now consider some properties of the autocorrelation function. It should quickly become apparent that not just any function of τ can be a valid autocorrelation function.

PROPERTY 1: The autocorrelation function evaluated at $\tau = 0$, $R_{XX}(0)$, is the *average normalized power* in the random process, $X(t)$.

To clarify this, note that $R_{XX}(0) = E[X^2(t)]$. Now suppose the random process $X(t)$ was a voltage measured at some point in a system. For a particular realization, $x(t)$, the instantaneous power would be $p(t) = x^2(t)/r$, where r is the impedance in Ohms

(Ω). The average power (averaged over all realizations in the ensemble) would then be $P_{\text{avg}} = E[X^2(t)]/r = R_{XX}(0)/r$. If, on the other hand, $X(t)$ were a current rather than a voltage, then the average power would be $P_{\text{avg}} = R_{XX}(0)r$. From a systems level, it is often desirable not to concern ourselves with whether a signal is a voltage or a current. Accordingly, it is common to speak of a *normalized power*, which is the power measured using a 1- Ω impedance. With $r = 1$, the two expressions for average power are the same and equal to the autocorrelation function evaluated at zero.

PROPERTY 2: The autocorrelation function of a WSS random process is an even function; that is, $R_{XX}(\tau) = R_{XX}(-\tau)$.

This property can easily be established from the definition of autocorrelation. Note that $R_{XX}(-\tau) = E[X(t)X(t - \tau)]$. Since $X(t)$ is WSS, this expression is the same for any value of t . In particular, replace t in the previous expression with $t + \tau$ so that $R_{XX}(-\tau) = E[X(t + \tau)X(t)] = R_{XX}(\tau)$. As a result of this property, any function of τ which is not even cannot be a valid autocorrelation function.

PROPERTY 3: The autocorrelation function of a WSS random process is maximum at the origin; that is, $|R_{XX}(\tau)| \leq R_{XX}(0)$ for all τ .

This property is established using the fact that for any two random variables, X and Y ,

$$(E[XY])^2 \leq E[X^2]E[Y^2]. \quad (78)$$

This fact was previously demonstrated in the proof of Theorem 5.4. Letting $X = X(t)$ and $Y = X(t + \tau)$ results in

$$R_{XX}^2(\tau) = \{E[X(t)X(t + \tau)]\}^2 \leq E[X^2(t)]E[X^2(t + \tau)] = R_{XX}^2(0). \quad (79)$$

Taking square roots of both sides results in Property 3.

PROPERTY 4: If $X(t)$ is ergodic and has no periodic components, then $\lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \mu_X^2$.

PROPERTY 5: If $X(t)$ has a periodic component, then $R_{XX}(t)$ will have a periodic component with the same period.

From these properties it is seen that an autocorrelation function can oscillate, can decay slowly or rapidly, and can have a nonzero constant component. As the name implies, the auto-correlation function is intended to measure the extent of correlation of samples of a random process as a function of how far apart the samples are taken.

1.16 Gaussian Random Processes

One of the most important classes of random processes is the Gaussian random process, which is defined as follows.

A random process, $X(t)$, for which any n samples, $X_1 = X(t_1)$, $X_2 = X(t_2)$, $\dots$, $X_n = X(t_n)$, taken at arbitrary points in time $t_1, t_2, \dots, t_n$, form a set of jointly Gaussian

random variables for any $n = 1, 2, 3, \dots$ is a Gaussian random process.

In vector notation, the vector of n samples, $\mathbf{X} = [X_1, X_2, \dots, X_n]^T$, will have a joint PDF given by

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det(C_{XX})}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_X)^T C_{XX}^{-1}(\mathbf{x} - \mu_X)\right). \quad (80)$$

As with any joint Gaussian PDF, all that is needed to specify the PDF is the mean vector and the covariance matrix. When the vector of random variables consists of samples of a random process, to specify the mean vector, all that is needed is the mean function of the random process, $\mu_X(t)$, since that will give the mean for any sample time. Similarly, all that is needed to specify the elements of the covariance matrix, $C_{ij} = \text{Cov}(X(t_i), X(t_j))$, would be the autocovariance function of the random process, $C_{XX}(t_1, t_2)$ or, equivalently, the autocorrelation function, $R_{XX}(t_1, t_2)$, together with the mean function. Hence, the mean and autocorrelation functions provide sufficient information to specify the joint PDF for any number of samples of a Gaussian random process. Note that since any n th order PDF is completely specified by $\mu_X(t)$ and $R_{XX}(t_1, t_2)$, if a Gaussian random process is WSS, then the mean and autocorrelation functions will be invariant to a time shift and hence any PDF will be invariant to a time shift. Therefore, any WSS Gaussian random process is also stationary in the strict sense.

If the process is zero-mean, the first order PDF is that of a zero-mean Gaussian random variable:

$$f_X(x; t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right). \quad (81)$$

This PDF is independent of time as would be expected for a stationary random process. Now consider the joint PDF of two samples, $X_1 = X(t)$ and $X_2 = X(t + \tau)$. Since the process is zero-mean, the mean vector is simply the all-zeros vector. The covariance matrix is then of the form

$$C_{XX} = \begin{bmatrix} R_{XX}(0) & R_{XX}(\tau) \\ R_{XX}(\tau) & R_{XX}(0) \end{bmatrix} = \sigma^2 \begin{bmatrix} 1 & \cos(\omega_0\tau) \\ \cos(\omega_0\tau) & 1 \end{bmatrix}. \quad (82)$$

The joint PDF of the two samples would then be

$$f_{X_1, X_2}(x_1, x_2; t, t + \tau) = \frac{1}{2\pi\sigma^2 |\sin(\omega_0\tau)|} \exp\left(-\frac{x_1^2 - 2x_1x_2 \cos(\omega_0\tau) + x_2^2}{2\sigma^2 \sin^2(\omega_0\tau)}\right). \quad (83)$$

Note once again that this joint PDF is dependent only on time difference, τ , and not on absolute time t . Higher order joint PDFs could be worked out in a similar manner.

1.17 Power Spectral Density

In the study of deterministic signals and systems, frequency domain techniques (e.g., Fourier transforms) provide a valuable tool that allows the engineer to gain significant insights into a variety of problems. In this chapter, we develop frequency domain tools for studying random processes. This will prepare us for the study of random processes in linear systems in the next chapter.

For a deterministic continuous signal, $x(t)$, the Fourier transform is used to describe its spectral content. In this text, we write the Fourier transform as

$$X(f) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt, \quad (84)$$

and the corresponding inverse transform is

$$x(t) = \mathcal{F}^{-1}[X(f)] = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft} df. \quad (85)$$

For discrete time signals, we could use a discrete Fourier transform (DFT) or a z -transform. The Fourier transform, $X(f)$, is referred to as the spectrum of $x(t)$ since it describes the spectral contents of the signal. In general, $X(f)$ is a complex function of frequency, and hence we also speak of an amplitude (magnitude) spectrum, $|X(f)|$, and a phase spectrum, $\angle X(f)$. In order to study random processes in the frequency domain, we seek a similar quantity that will describe the spectral characteristics of a random process.

The most obvious thing to do would be to try to define the Fourier transform of a random process as perhaps

$$X(f) = \int_{-\infty}^{\infty} X(t)e^{-j2\pi ft} dt = \mathcal{F}[X(t)]; \quad (86)$$

however, this leads to several problems. First of all, there are problems with existence. Since $X(t)$ is a random process, there is not necessarily any guarantee that the integral exists for every possible realization, $x(t)$. That is, not every realization of the random process may have a Fourier transform. Even for processes that are well-behaved in the sense that every realization has a well-defined Fourier transform, we are still left with the problem that $X(f)$ is itself a random process.

1.18 Definition of Power Spectral Density

To start with, for a random process $X(t)$, define a truncated version of the random process as

$$X_{t_0}(t) = \begin{cases} X(t), & |t| \leq t_0 \\ 0, & |t| > t_0 \end{cases} \quad (87)$$

The energy of this random process is

$$E_{X_{t_0}} = \int_{-t_0}^{t_0} X^2(t) dt = \int_{-\infty}^{\infty} X_{t_0}^2(t) dt, \quad (88)$$

and hence the time averaged power is

$$P_{X_{t_0}} = \frac{1}{2t_0} \int_{-\infty}^{\infty} X_{t_0}^2(t) dt = \frac{1}{2t_0} \int_{-\infty}^{\infty} |X_{t_0}(f)|^2 df. \quad (89)$$

The last equality is obtained using Parseval's theorem. The quantity $X_{t_0}(f)$ is the Fourier transform of $X_{t_0}(t)$. Since the random process has been truncated to a finite time interval, there will not be any problem with the existence of the Fourier transform. Note that $P_{X_{t_0}}$ is a random variable and so to get the ensemble averaged power, we must take an expectation,

$$\overline{P}_{X_{t_0}} = E[P_{X_{t_0}}] = \frac{1}{2t_0} \int_{-\infty}^{\infty} E[|X_{t_0}(f)|^2] df. \quad (90)$$

The power in the (untruncated) random process $X(t)$ is then found by passing to the limit as $t_0 \rightarrow \infty$,

$$\overline{P}_X = \lim_{t_0 \rightarrow \infty} \frac{1}{2t_0} \int_{-\infty}^{\infty} E[|X_{t_0}(f)|^2] df = \int_{-\infty}^{\infty} \lim_{t_0 \rightarrow \infty} \frac{E[|X_{t_0}(f)|^2]}{2t_0} df. \quad (91)$$

Define $S_{XX}(f)$ to be the integrand in the above equation. That is,

$$S_{XX}(f) = \lim_{t_0 \rightarrow \infty} \frac{E[|X_{t_0}(f)|^2]}{2t_0}. \quad (92)$$

Then, the average power in the process can be expressed as

$$\overline{P}_X = \int_{-\infty}^{\infty} S_{XX}(f) df. \quad (93)$$

Hence, this function of frequency which we have referred to simply as $S_{XX}(f)$ has the property that when integrated over all frequency, the total power in the process is obtained. In other words, $S_{XX}(f)$ has the units of power per unit frequency and so it is the power density function of the random process in the frequency domain. Thus, the quantity $S_{XX}(f)$ is given the name power spectral density (PSD). In summary, we have the following definition of PSD.

1.18.1 Definition

For a random process $X(t)$, the power spectral density (PSD) is defined as

$$S_{XX}(f) = \lim_{t_0 \rightarrow \infty} \frac{E[|X_{t_0}(f)|^2]}{2t_0}, \quad (94)$$

where $X_{t_0}(f)$ is the Fourier transform of the truncated version of the process.

Several properties of the PSD function should be evident from this definition and the development:

1. $S_{XX}(f)$ is a real function.
2. $S_{XX}(f)$ is a nonnegative function.
3. $S_{XX}(f)$ is an even function.
4. The average power is given by

$$\overline{P}_X = \int_{-\infty}^{\infty} S_{XX}(f) df. \quad (95)$$

1.18.2 Definition

The cross spectral density between two random processes $X(t)$ and $Y(t)$ is the Fourier transform of the cross-correlation function:

$$S_{XY}(f) = \mathcal{F}[R_{XY}(\tau)] = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j2\pi f\tau} d\tau. \quad (96)$$

The cross spectral density does not have a physical interpretation nor does it share the same properties as the PSD function. For example, $S_{XY}(f)$ is not necessarily real since $R_{XY}(\tau)$ is not necessarily even. The cross spectral density function does possess a form of symmetry known as Hermitian symmetry,

$$S_{XY}(f) = S_{YX}(-f) = S_{XY}^*(-f). \quad (97)$$

This property follows from the fact that $R_{XY}(\tau) = R_{YX}(-\tau)$.

1.19 Wiener–Khinchine Relations

The autocorrelation function $R_{XX}(\tau)$ and the power spectral density $S_{XX}(f)$ form a Fourier transform pair.

$$S_{XX}(f) = \int_{-\infty}^{\infty} R_{XX}(\tau) \exp(-j2\pi f\tau) d\tau \quad (133)$$

$$R_{XX}(\tau) = \int_{-\infty}^{\infty} S_{XX}(f) \exp(j2\pi f\tau) df \quad (134)$$

For practical application purposes, the power spectral density is the more useful function of the two relations above.

1.20 Power of a Random Process

The power P_X (average power) of a wide-sense stationary random process $x(t)$ is its mean square value $\overline{x^2}$.

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df \quad (135)$$

$$P_X = \overline{X^2} = R_X(0) = \int_{-\infty}^{\infty} S_X(f) df \quad (136)$$

Because $S_X(f)$ is an even function of f , we have

$$P_X = \overline{X^2} = 2 \int_0^{\infty} S_X(f) df \quad (137)$$

Thus,

$P_X \rightarrow$ the area under the PSD.

$P_X = \overline{X^2} \rightarrow$ the ensemble mean of the squared amplitudes of the sample functions at any instant.

- The PSD may not exist for processes that are not wide-sense stationary.

1.20.1 Example

Determine the autocorrelation function $R_X(\tau)$ and the power P_X of a low-pass random process with a white noise PSD

$$S_X(f) = \frac{N_0}{2}.$$

Solution

The given power spectral density is band-limited to $|f| \leq B$ and can be written as

$$S_X(f) = \frac{N_0}{2} \Pi\left(\frac{f}{2B}\right)$$

where $\Pi(\cdot)$ is the rectangular function.

Using the inverse Fourier transform relation,

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df$$

$$R_X(\tau) = \int_{-B}^B \frac{N_0}{2} e^{j2\pi f\tau} df$$

$$R_X(\tau) = \frac{N_0}{2} \left[\frac{e^{j2\pi f\tau}}{j2\pi\tau} \right]_{-B}^B$$

$$R_X(\tau) = N_0 B \operatorname{sinc}(2\pi B\tau)$$

Hence, the autocorrelation function is

$$\boxed{R_X(\tau) = N_0 B \operatorname{sinc}(2\pi B\tau)}$$

The power of the process is

$$P_X = \overline{x^2} = R_X(0)$$

$$P_X = N_0 B$$

Alternatively,

$$P_X = 2 \int_0^\infty S_X(f) df = 2 \int_0^B \frac{N_0}{2} df = N_0 B$$

$$\boxed{P_X = N_0 B}$$

1.20.2 Multiple Random Processes

For two real random processes $x(t)$ and $y(t)$, we define the **cross-correlation function** as

$$R_{xy}(t_1, t_2) = \overline{x(t_1) y(t_2)}.$$

The two processes are said to be **jointly stationary** if each of the processes is individually wide-sense stationary and if

$$R_{xy}(t_1, t_2) = R_{xy}(t_2 - t_1) = R_{xy}(\tau)$$

Uncorrelated, Orthogonal and Independent Processes

Two processes $x(t)$ and $y(t)$ are said to be **uncorrelated** if their cross-correlation function is equal to the product of their means, that is,

$$R_{xy}(\tau) = \overline{x(t) y(t + \tau)} = \bar{x} \bar{y}.$$

Hence, $x(t)$ and $y(t + \tau)$ are uncorrelated for all t and τ .

The processes $x(t)$ and $y(t)$ are said to be **orthogonal** if

$$R_{xy}(\tau) = 0.$$

The processes $x(t)$ and $y(t)$ are said to be **independent random processes** if the random variables $x(t_1)$ and $y(t_2)$ are independent for all possible choices of t_1 and t_2 .

Sum of Random Processes

If two stationary processes $x(t)$ and $y(t)$ are added to form a process $z(t)$, the statistics of $z(t)$ can be determined in terms of those of $x(t)$ and $y(t)$.

If

$$z(t) = x(t) + y(t)$$

then

$$\begin{aligned} R_z(\tau) &= \overline{z(t)z(t+\tau)} \\ &= \overline{(x(t) + y(t))(x(t+\tau) + y(t+\tau))} \\ &= R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau) \end{aligned}$$

If $x(t)$ and $y(t)$ are uncorrelated, then

$$R_{XY}(\tau) = R_{YX}(\tau) = \bar{X} \bar{Y}$$

and

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau) + 2\bar{X}\bar{Y}$$

If $x(t)$ and $y(t)$ are uncorrelated with zero means,

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau)$$

$$S_Z(f) = S_X(f) + S_Y(f)$$

$$\overline{Z^2} = \overline{X^2} + \overline{Y^2}$$

The mean square value of a sum of orthogonal or incoherent processes is equal to the sum of the mean square values of the individual processes.

Multiplication of Random Processes

Consider the case when two independent, jointly wide-sense stationary random processes are multiplied together.

$$W(t) = X(t)Y(t)$$

The mean of $w(t)$ is

$$E[W(t)] = \mu_W = E[X(t)]E[Y(t)] = \mu_X\mu_Y$$

The variance of $w(t)$ is

$$\sigma_W^2 = E[(X(t)Y(t) - \mu_X\mu_Y)^2]$$

$$= (\sigma_X^2 + \mu_X^2)(\sigma_Y^2 + \mu_Y^2) - \mu_X^2\mu_Y^2$$

The autocorrelation function of $w(t)$ is

$$R_W(\tau) = E[X(t)Y(t)X(t+\tau)Y(t+\tau)]$$

$$= E[X(t)X(t+\tau)]E[Y(t)Y(t+\tau)]$$

$$= R_X(\tau)R_Y(\tau)$$

Thus, for independent WSS processes, the autocorrelation of the product is the product of the autocorrelations.

1.21 Bandwidth of a Random Process

Now that we have an analytical function that describes the spectral content of a signal, it is appropriate to talk about the bandwidth of a random process. As with deterministic signals, there are many definitions of bandwidth. Which definition is used depends on the application and sometimes on personal preference. Several definitions of bandwidth are given next. To understand these definitions, it is helpful to remember that when measuring the bandwidth of a signal (whether random or deterministic), only positive frequencies are measured. Also, we tend to classify signals according to where their spectral contents lies. Those signals for which most of the power is at or near direct current (d.c.) are referred to as *lowpass* signals, while those signals whose PSD is centered around some nonzero frequency, $f = f_0$, are referred to as *bandpass* processes.

1.21.1 Definition

For a lowpass process, the absolute bandwidth, B_{abs} , is the largest frequency for which the PSD is nonzero. That is, B_{abs} is the smallest value of B such that $S_{XX}(f) = 0$ for all

$f > B$. For a bandpass process, let B_L be the largest value of B such that $S_{XX}(f) = 0$ for all $0 < f < B$, and similarly let B_R be the smallest value of B such that $S_{XX}(f) = 0$ for all $B < f$. Then $B_{\text{abs}} = B_R - B_L$. In summary, the absolute bandwidth of a random process is the width of the band that contains all frequency components. The concept of absolute bandwidth is illustrated in Figure 2.

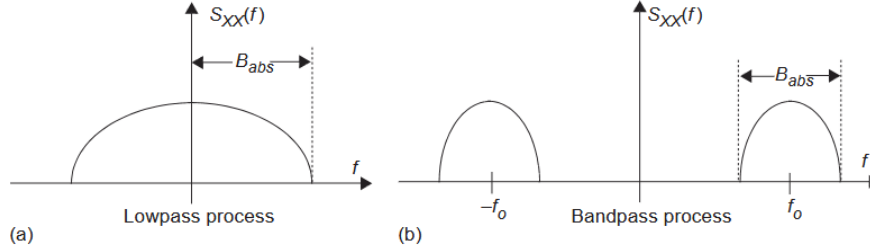


Figure 2: Measuring the absolute bandwidth of a lowpass (a) and bandpass (b) process

1.21.2 Definition

The 3-dB bandwidth (or half power bandwidth), $B_{3\text{dB}}$, is the width of the frequency band where the PSD is within 3 dB of its peak value everywhere within the band. Let S_{peak} be the maximum value of the PSD. Then for a lowpass signal, $B_{3\text{dB}}$ is the largest value of B for which $S_{XX}(f) > S_{\text{peak}}/2$ for all frequencies such that $0 < f < B$. For a bandpass process, $B_{3\text{dB}} = B_R - B_L$, where $S_{XX}(f) > S_{\text{peak}}/2$ for all frequencies such that $B_L < f < B_R$ and it is assumed that the peak value occurs within the band. The concept of 3-dB bandwidth is illustrated in Figure 3.

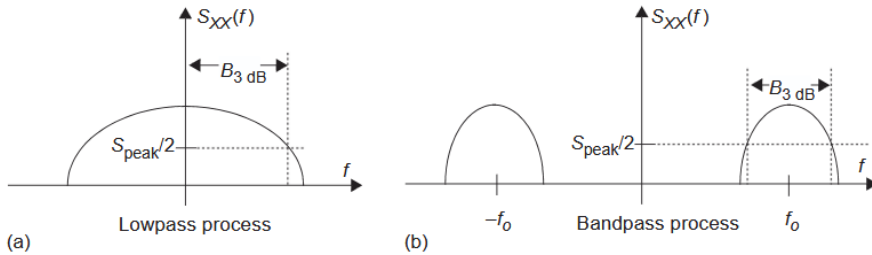


Figure 3: Measuring the 3-dB bandwidth of a lowpass (a) and bandpass (b) process

1.22 Continuous Time Linear Systems

Consider an LTI system described by an impulse response $h(t)$ or a transfer function $H(f)$. If a random process, $X(t)$, is input to this system, the output will also be random and is given by the convolution integral

$$Y(t) = \int_{-\infty}^{\infty} X(u)h(t-u) du. \quad (98)$$

We would like to statistically describe the output of the system. Ultimately, the joint PDF of any number of samples of the output would be nice. In general, this is a very difficult problem and hence we have to be satisfied with a simpler description. However, if the input process is Gaussian, then the output process will also be Gaussian since any linear processing of Gaussian random variables (processes) produces new Gaussian random variables (processes). In that case, to completely describe the output of the system, we need merely to compute the mean and the autocovariance (or autocorrelation) function of the output. Even if the processes involved are not Gaussian, the mean and autocorrelation functions will serve as a good start toward describing the process. Hence, our first goal will be to specify the mean and autocorrelation functions of the output of an LTI system with a random input.

$$R_{YY}(\tau) = R_{XX}(\tau) * h(\tau) * h(-\tau). \quad (99)$$

Hence, the output autocorrelation function is found by a double convolution. The presence of the double convolution in the above equation suggests an equivalent frequency domain representation. Taking Fourier transforms of both sides gives an expression for the power spectral density (PSD) of the output of the filter in terms of the input PSD:

$$S_{YY}(f) = S_{XX}(f)H(f)H^*(f) = S_{XX}(f)|H(f)|^2. \quad (100)$$

The term $|H(f)|^2$ is sometimes referred to as the *power transfer function* because it describes how the power is transferred from the input to the output of the system. In summary, we have shown the following results.

1.22.1 Example

White Gaussian noise, $N(t)$ with a PSD of $S_{NN}(f) = \frac{N_0}{2}$ is input to a resistor, capacitor (RC) lowpass filter. Such a filter will have a transfer function and impulse response given by

$$H(f) = \frac{1}{1 + j2\pi fRC} \quad (101)$$

and

$$h(t) = \frac{1}{RC} \exp\left(-\frac{t}{RC}\right) u(t), \quad (102)$$

respectively. If the input noise is zero-mean, $\mu_N = 0$, then the output process will also be zero-mean, $\mu_Y = 0$. Also,

$$S_{YY}(f) = S_{NN}(f)|H(f)|^2 = \frac{N_0/2}{1 + (2\pi fRC)^2}. \quad (103)$$

Using inverse Fourier transforms, the output autocorrelation is found to be

$$R_{YY}(\tau) = \frac{N_0}{4RC} \exp\left(-\frac{|\tau|}{RC}\right). \quad (104)$$

1.23 Noise Equivalent Bandwidth

Consider an ideal lowpass filter with a bandwidth B whose transfer function is shown in Figure 4. Suppose white Gaussian noise with PSD $N_0/2$ is passed through this filter. The total output power would be $P_o = N_0 B$. For an arbitrary lowpass filter, the output noise power would be

$$P_o = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df. \quad (105)$$

One way to define the bandwidth of an arbitrary filter is to construct an ideal lowpass filter that produces the same output power as the actual filter. This results in the following definition of bandwidth known as *noise equivalent bandwidth*.

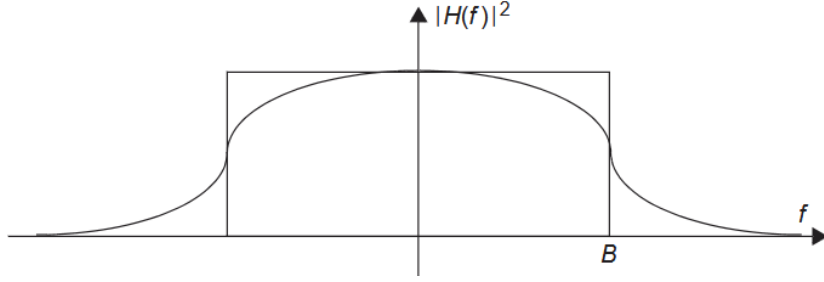


Figure 4: Power transfer function of an arbitrary and ideal lowpass filter. $B = B_{\text{neq}}$ if areas under the two curves are equal

1.24 Signal-to-Noise Ratios

Often the input to a linear system will consist of signal plus noise, namely,

$$X(t) = S(t) + N(t), \quad (106)$$

where the signal part can be deterministic or a random process. We can invoke linearity to show that the mean process of the output can be viewed as a sum of the mean due to the signal input alone plus the mean due to the noise input alone. That is,

$$\mu_Y(t) = \mu_S(t) * h(t) + \mu_N(t) * h(t). \quad (107)$$

In most cases, the noise is taken to be zero-mean, in which case the mean at the output is due to the signal part alone.

When calculating the autocorrelation function of the output, we cannot invoke superposition since autocorrelation is not a linear operation. First, we calculate the autocorrelation function of the signal plus noise input.

$$R_{XX}(t_1, t_2) = E[(S(t_1) + N(t_1))(S(t_2) + N(t_2))]. \quad (108)$$

If the signal and noise part are independent, which is generally a reasonable assumption, and the noise is zero-mean, then this autocorrelation becomes

$$R_{XX}(t_1, t_2) = R_{SS}(t_1, t_2) + R_{NN}(t_1, t_2), \quad (109)$$

or, assuming all processes involved are WSS,

$$R_{XX}(\tau) = R_{SS}(\tau) + R_{NN}(\tau). \quad (110)$$

As a result, the PSD of the output can be written as

$$S_{YY}(f) = S_{XX}(f)|H(f)|^2 = S_{SS}(f)|H(f)|^2 + S_{NN}(f)|H(f)|^2, \quad (111)$$

which is composed of two terms, namely that due to the signal and that due to the noise. We can then calculate the output power due to the signal part and the output power due to the noise part.

1.24.1 Definition

The signal-to-noise ratio (SNR) for a signal comprised of the sum of a desired (signal) part and a noise part is defined as the ratio of the power of the signal part to the power (variance) of the noise part. That is, for $X(t) = S(t) + N(t)$,

$$\text{SNR} = \frac{E[S^2(t)]}{E[N^2(t)]} = \frac{R_{SS}(0)}{R_{NN}(0)} = \frac{\int_{-\infty}^{\infty} S_{SS}(f) df}{\int_{-\infty}^{\infty} S_{NN}(f) df}. \quad (112)$$

Example 1

Consider the sinusoidal random process

$$x(t) = A \sin(\omega_0 t)$$

where A is a uniform random variable over $(0, 1)$.

Find the mean and autocorrelation function. Check whether the process is WSS.

Solution

$$\mu_x(t) = E[x(t)] = E[A] \sin(\omega_0 t)$$

$$E[A] = \int_0^1 A \, dA = \frac{1}{2}$$

$$\mu_x(t) = \frac{1}{2} \sin(\omega_0 t)$$

Since the mean depends on time, the process is not stationary.

The autocorrelation function is

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$$

$$= E[A^2 \sin(\omega_0 t_1) \sin(\omega_0 t_2)]$$

$$E[A^2] = \int_0^1 A^2 \sin(\omega_0 t_1) \sin(\omega_0 t_2) dA = \frac{1}{3} \sin(\omega_0 t) \sin(\omega_0(t + \tau))$$

$$R_x(t_1, t_2) = \frac{1}{3} \sin(\omega_0 t_1) \sin(\omega_0 t_2)$$

Hence, R_x depends on t_1 and t_2 , so the process is not WSS.

Example 2

Consider a sinusoid with F uniformly distributed over $(0, f_0)$.

$$x(t) = \cos(2\pi Ft), \quad F \rightarrow \text{RV}$$

Find the mean function.

Solution:

$$\begin{aligned}
\mu_{x(t)} &= \mathbb{E}[\cos(2\pi Ft)] = \frac{1}{f_0} \int_0^{f_0} \cos(2\pi ft) df \\
&= \frac{\sin(2\pi f_0 t)}{2\pi f_0 t} \\
&= \text{sinc}(2f_0 t)
\end{aligned}$$

Example 3

Show that the random process

$$x(t) = A \cos(\omega_c t + \theta)$$

is wide-sense stationary, where θ is uniformly distributed over $(0, 2\pi)$.

Solution

$$\begin{aligned}
E[x(t)] &= AE[\cos(\omega_c t + \theta)] \\
&= A \int_0^{2\pi} \cos(\omega_c t + \theta) P_\theta(\theta) d\theta = 0 \\
&= \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega_c t + \theta) d\theta = 0
\end{aligned}$$

The autocorrelation function is

$$\begin{aligned}
R_x(t_1, t_2) &= E[x(t_1)x(t_2)] \\
&= A^2 E[\cos(\omega_c t_1 + \theta) \cos(\omega_c t_2 + \theta)] \\
&= \frac{A^2}{2} \cos[\omega_c(t_2 - t_1)]
\end{aligned}$$

Let $\tau = t_2 - t_1$,

$$R_x(\tau) = \frac{A^2}{2} \cos(\omega_c \tau)$$

Thus, the process is WSS.

Example 4

Let $X(t)$ and $Y(t)$ be zero-mean and WSS random processes.

Consider the random process $Z(t) = X(t) + Y(t)$

- (a) Determine the ACF and power spectral density of $Z(t)$ if $X(t)$ and $Y(t)$ are jointly WSS.
- (b) Repeat part (a) if $X(t)$ and $Y(t)$ are orthogonal.
- (c) If $X(t)$ and $Y(t)$ are orthogonal, find the mean square value of $Z(t)$.

Solution:

$$R_{ZZ}(t_1, t_2) = \mathbb{E}[Z(t_1)Z(t_2)]$$

$$= \mathbb{E}[(X(t_1) + Y(t_1))(X(t_2) + Y(t_2))]$$

$$= \mathbb{E}[X(t_1)X(t_2) + X(t_1)Y(t_2) + Y(t_1)X(t_2) + Y(t_1)Y(t_2)]$$

$$= R_{XX}(t_1, t_2) + R_{XY}(t_1, t_2) + R_{YX}(t_1, t_2) + R_{YY}(t_1, t_2)$$

Example 5

Consider

$$x(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

where A and B are independent zero-mean Gaussian random variables with equal variance σ^2 .

Solution

$$\mu_x(t) = E[A] \cos(\omega_0 t) + E[B] \sin(\omega_0 t) = 0$$

$$R_x(t_1, t_2) = E[x(t_1)x(t_2)]$$

$$= \sigma^2 \cos[\omega_0(\tau)]$$

Thus, the process is WSS.

The first-order PDF is

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

White Noise Filtering

White noise has infinite power and cannot be sampled directly. Hence, it must be filtered first.

Let $N_f(t)$ be the output of an ideal low-pass filter with bandwidth B .

$$S_{N_f}(f) = \begin{cases} \frac{N_0}{2}, & |f| < B \\ 0, & \text{otherwise} \end{cases}$$

The autocorrelation function is

$$R_{N_f}(\tau) = N_0 B \operatorname{sinc}(2B\tau)$$

Example 6

Determine the PSD and mean square value of

$$x(t) = A \cos(\omega_c t + \theta)$$

where θ is uniformly distributed over $(0, 2\pi)$.

Solution

$$R_x(\tau) = \frac{A^2}{2} \cos(\omega_c \tau)$$

$$S_x(f) = \frac{A^2}{4} [\delta(f - f_c) + \delta(f + f_c)]$$

$$P_x = \overline{x^2} = R_x(0) = \frac{A^2}{2}$$

Example 7

Determine the autocorrelation function and the PSD of the DSB-SC modulated process $m(t) \cos(\omega_c t + \theta)$, where $m(t)$ is a wide-sense stationary random process and θ is a random variable uniformly distributed over $(0, 2\pi)$ and independent of $m(t)$.

Sol:

Let

$$\phi(t) = m(t) \cos(\omega_c t + \theta)$$

The autocorrelation function is

$$R_\phi(\tau) = \mathbb{E}[m(t) \cos(\omega_c t + \theta) m(t + \tau) \cos(\omega_c(t + \tau) + \theta)]$$

Since $m(t)$ and θ are independent,

$$R_\phi(\tau) = \mathbb{E}[m(t)m(t + \tau)] \mathbb{E}[\cos(\omega_c t + \theta) \cos(\omega_c(t + \tau) + \theta)]$$

Using the identity:

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

and averaging over θ , we get

$$R_\phi(\tau) = \frac{1}{2}R_m(\tau) \cos(\omega_c \tau)$$

The power spectral density is given by the Fourier transform of $R_\phi(\tau)$:

$$S_\phi(f) = \frac{1}{4} [S_m(f + f_c) + S_m(f - f_c)]$$

The power of the modulated signal is

$$\sigma_\phi^2 = R_\phi(0) = \frac{1}{2}R_m(0) = \frac{1}{2}\sigma_m^2$$

Hence, the power of the DSB-SC modulated signal is half the power of the modulating signal.